

SIMATS ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING ASSESSMENT TOOL 3: Comparative Analysis Solutions

COURSE NAME: COMPUTER VISION	COURSE CODE: ITA05
COURSE OUTCOME: CO4 (BTL 5)	WEIGHTAGE: 10%

Question 15: Recognition Precision Comparison

Given Data:

- System A: TP = 90, FP = 10
- System B: TP = 85, FP = 5

(a) Precision Calculation:

Formula: Precision = True Positives / (True Positives + False Positives)

System A Precision:

- Precision_A = $90 / (90 + 10) = 90 / 100 = 0.90$ (90.0%)

System B Precision:

- Precision_B = $85 / (85 + 5) = 85 / 90 = 0.9444$ (94.44%)

(b) Comparison and Evaluation:

- Comparison: System B achieves higher precision (94.44%) compared to System A (90.00%).
- Evaluation & Justification: Precision measures the accuracy of positive predictions (i.e., how many of the identified targets are correct). System B incurs significantly fewer false positives (5 vs 10), making it more reliable in applications where false alarms are costly (e.g., medical diagnosis or automated surveillance). Therefore, System B performs better in terms of precision.

Question 10: Error Rate Comparison

Given Data:

- Method A: Errors = 20, Total = 100
- Method B: Errors = 10, Total = 100

(a) Error Rate Calculation:

Formula: Error Rate = Total Errors / Total Samples

Method A Error Rate:

- Error Rate_A = $20 / 100 = 0.20$ (20.0%)

Method B Error Rate:

- Error Rate_B = $10 / 100 = 0.10$ (10.0%)

(b) Comparison and Evaluation:

- Comparison: Method B has an error rate of 10.0%, which is exactly half the error rate of Method A (20.0%).
- Evaluation & Justification: A lower error rate directly translates to higher overall classification/processing correctness (Method B accuracy = 90%, Method A accuracy = 80%). Consequently, Method B is significantly superior and more dependable.

Question 4: Depth Estimation Comparison (Stereo vs SfM)

Given Data:

- Actual Depths: Object A = 2.0 m, Object B = 3.0 m, Object C = 4.0 m
- Stereo Vision Depths: A = 2.0 m, B = 3.0 m, C = 4.0 m
- Structure from Motion (SfM) Depths: A = 2.5 m, B = 3.8 m, C = 5.0 m

(a) Absolute Error Calculation:

Formula: Absolute Error = | Estimated Depth - Actual Depth |

Stereo Vision Absolute Errors:

- Object A: $|2.0 - 2.0| = 0.0$ m
- Object B: $|3.0 - 3.0| = 0.0$ m
- Object C: $|4.0 - 4.0| = 0.0$ m
- Mean Absolute Error (MAE_Stereo) = $(0.0 + 0.0 + 0.0) / 3 = 0.0$ m

Structure from Motion (SfM) Absolute Errors:

- Object A: $|2.5 - 2.0| = 0.5$ m
- Object B: $|3.8 - 3.0| = 0.8$ m
- Object C: $|5.0 - 4.0| = 1.0$ m
- Mean Absolute Error (MAE_SfM) = $(0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 \approx 0.767$ m

(b) Comparison and Evaluation:

- Comparison: Stereo Vision achieves an absolute error of 0.0 m across all objects, whereas SfM exhibits an average absolute error of 0.767 m.
- Evaluation & Justification: Stereo Vision demonstrates perfect accuracy for the tested scene. SfM suffers from metric scale ambiguity inherent in monocular sequence reconstruction, leading to a systematic overestimation of depth. Thus, Stereo Vision is far more accurate for exact metric depth recovery.

Question 3: Optical Flow Magnitude Comparison

Given Data (Optical Flow Magnitudes in pixels/frame):

- Region R1: Method A = 5, Method B = 8
- Region R2: Method A = 6, Method B = 10
- Region R3: Method A = 5, Method B = 9

(a) Average Motion Magnitude Calculation:

Formula: Average Magnitude = Sum of Magnitudes across Regions / Total Regions

Method A Average Magnitude:

- $\text{Avg_A} = (5 + 6 + 5) / 3 = 16 / 3 \approx 5.33 \text{ pixels/frame}$

Method B Average Magnitude:

- $\text{Avg_B} = (8 + 10 + 9) / 3 = 27 / 3 = 9.00 \text{ pixels/frame}$

(b) Comparison and Evaluation:

- Comparison: Method B yields a significantly higher average optical flow magnitude (9.00 px/frame) compared to Method A (5.33 px/frame).
- Evaluation & Justification: Method B captures stronger motion vectors across all regions. Higher magnitude capture indicates better sensitivity to fast-moving objects or large pixel displacements, making Method B better suited for high-dynamic scenes.

Question 7: F1-Score Comparison

Given Data:

- Model A: Precision = 0.8, Recall = 0.6
- Model B: Precision = 0.75, Recall = 0.75

(a) F1-Score Calculation:

Formula: $\text{F1-Score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$

Model A F1-Score:

- $\text{F1_A} = 2 * (0.8 * 0.6) / (0.8 + 0.6) = 2 * 0.48 / 1.4 = 0.96 / 1.4 \approx 0.6857 \text{ (68.57\%)}$

Model B F1-Score:

- $\text{F1_B} = 2 * (0.75 * 0.75) / (0.75 + 0.75) = 2 * 0.5625 / 1.5 = 1.125 / 1.5 = 0.7500 \text{ (75.00\%)}$

(b) Comparison and Evaluation:

- Comparison: Model B has an F1-score of 0.75, which is higher than Model A's F1-score of 0.6857.
- Evaluation & Justification: The F1-score represents the harmonic mean of precision and recall. Model A suffers from a lower recall (0.6), which penalizes its harmonic mean. Model B provides a perfectly balanced trade-off between precision and recall, resulting in a superior overall performance metric. Thus, Model B is better.

Question 1: Detection Performance Comparison

Given Data:

- Model A (Viola-Jones): TP = 80, FP = 20, FN = 30
- Model B (YOLO): TP = 90, FP = 15, FN = 20

(a) Precision and Recall Calculation:

Formulas:

- **Precision = $\text{TP} / (\text{TP} + \text{FP})$**
- **Recall = $\text{TP} / (\text{TP} + \text{FN})$**

Model A (Viola-Jones):

- $\text{Precision_A} = 80 / (80 + 20) = 80 / 100 = 0.80 \text{ (80.0\%)}$

- $\text{Recall_A} = 80 / (80 + 30) = 80 / 110 \approx 0.7273$ (72.73%)

Model B (YOLO):

- $\text{Precision_B} = 90 / (90 + 15) = 90 / 105 \approx 0.8571$ (85.71%)
- $\text{Recall_B} = 90 / (90 + 20) = 90 / 110 \approx 0.8182$ (81.82%)

(b) Comparison and Evaluation:

- Comparison: Model B outperforms Model A in both Precision (85.71% vs 80.00%) and Recall (81.82% vs 72.73%).
- Evaluation & Justification: YOLO (Model B) generates fewer false positives and misses fewer actual objects (fewer false negatives). Consequently, Model B is clearly superior for modern object detection tasks.

Question 12: IOU Comparison

Given Data (IOU values):

- Object 1: Model A = 0.60, Model B = 0.80
- Object 2: Model A = 0.70, Model B = 0.85
- Object 3: Model A = 0.65, Model B = 0.90

(a) Average IOU Calculation:

Formula: Average IOU = Sum of IOU values / Number of Objects

Model A Average IOU:

- $\text{Avg IOU_A} = (0.60 + 0.70 + 0.65) / 3 = 1.95 / 3 = 0.65$ (65.0%)

Model B Average IOU:

- $\text{Avg IOU_B} = (0.80 + 0.85 + 0.90) / 3 = 2.55 / 3 = 0.85$ (85.0%)

(b) Comparison and Evaluation:

- Comparison: Model B achieves a significantly higher mean IOU (0.85) than Model A (0.65).
- Evaluation & Justification: Intersection over Union (IOU) quantifies the overlap between predicted and ground-truth bounding boxes. A higher average IOU indicates far tighter and more accurate bounding box localization. Thus, Model B provides superior detection quality.

Question 9: Detection Accuracy Comparison

Given Data:

- Model A: Correct = 180, Total = 200
- Model B: Correct = 170, Total = 200

(a) Accuracy Calculation:

Formula: Accuracy = Correct Predictions / Total Samples

Model A Accuracy:

- $\text{Accuracy_A} = 180 / 200 = 0.90$ (90.0%)

Model B Accuracy:
• $\text{Accuracy}_B = 170 / 200 = 0.85 \text{ (85.0\%)}$

- (b) Comparison and Evaluation:**
- Comparison: Model A achieves 90.0% accuracy, while Model B achieves 85.0% accuracy.
 - Evaluation & Justification: Model A correctly identifies 10 more samples out of 200 than Model B, demonstrating superior overall performance and higher reliability.

Summary Comparison Table

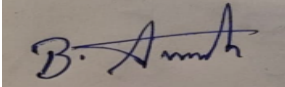
Question / Topic	Calculated Results	Better Technique	Key Justification
Q15: Recognition Precision	System A: 90.00% System B: 94.44%	System B	Higher precision (fewer false positives)
Q10: Error Rate	Method A: 20.00% Method B: 10.00%	Method B	Half the error rate of Method A
Q4: Depth Estimation Error	Stereo MAE: 0.00 m SfM MAE: 0.767 m	Stereo Vision	Zero metric error; SfM suffers from scale ambiguity
Q3: Optical Flow Magnitude	Method A: 5.33 px/frame Method B: 9.00 px/frame	Method B	Captures stronger motion vectors
Q7: F1-Score	Model A: 0.6857 Model B: 0.7500	Model B	Better balance between precision and recall
Q1: Detection Performance	Model A: P=80.0%, R=72.7% Model B: P=85.7%, R=81.8%	Model B (YOLO)	Superior precision and recall overall
Q12: IOU Comparison	Model A Mean: 0.65 Model B Mean: 0.85	Model B	Significantly tighter bounding box localization
Q9: Detection Accuracy	Model A: 90.00% Model B: 85.00%	Model A	Higher correct classification rate (180/200)

Code of Conduct

I Avinash B (192511193) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A handwritten signature in black ink, appearing to read "B. Avinash", is written on a light-colored rectangular background.